

Original Article

Impact of Financial Literacy on Financial Performance in Select Multi-Purpose Cooperatives in Tagbilaran City, Bohol, Philippines

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Abstract. Cooperatives play a vital role in promoting economic growth and financial inclusion, especially in communities with limited access to formal financial institutions. Despite this importance, many cooperative members still exhibit low financial literacy, which may affect their financial decisions and the organization's overall performance. This study utilized a quantitative descriptive-correlational research design with mediation analysis involving one hundred members from four registered multi-purpose cooperatives. Data on members' financial literacy were gathered through a questionnaire, while financial performance indicators were obtained from cooperative records. Results revealed that financial knowledge significantly increases financial attitude ($\beta = 0.525$, $p < .001$), and financial attitude significantly increases financial behavior ($\beta = 0.592$, $p < .001$). However, financial knowledge has no significant direct effect on financial behavior ($\beta = 0.024$, $p = .797$), demonstrating that financial attitude fully mediates this pathway (indirect effect $\beta = 0.311$, $p < .001$). Furthermore, regression analysis showed that members' financial literacy had no significant predictive effect on cooperative financial performance ($\beta = 0.048$, $p = .632$), with the model explaining only 0.2% of the variance ($R^2 = 0.002$). These findings indicate that improving members' financial literacy alone is insufficient to enhance cooperative outcomes; organizational factors such as governance, management, and internal controls likely exert a more substantial impact.

Keywords: Cooperatives; Financial literacy; Financial performance; Multi-purpose cooperatives

Cooperatives are more than just businesses; they are vital engines of economic development and social cohesion, particularly in regions where formal financial services are scarce. In Bohol, a landscape shaped by agriculture, tourism, and micro-enterprises, cooperatives serve as a financial lifeline. For instance, the Bohol Community Multi-Purpose Cooperative reports that 40% of its members are farmers and fisherfolk, a demographic that requires specialized savings and lending solutions. As member-owned entities, these organizations' structural strength is inherently tied to the financial competence of their constituent members.

This study is grounded in Human Capital Theory, which posits that investments in knowledge and skills directly enhance economic productivity. In the context of a cooperative, a member's financial literacy is not merely a

personal asset but a form of collective human capital that drives the organization's operational success. Financial literacy serves as the essential structural link between individual behavior and institutional health (Perez & Lopez, 2020). Viewed through the lens of the Resource-Based View (RBV) theory, prominent literacy gaps represent a clear failure to leverage internal human resources – the members themselves – which are necessary for building a competitive, sustainable organization.

However, a significant gap remains between this ideal and global reality. The Economic and Social Survey of Asia and the Pacific (2022) highlights an ongoing literacy crisis, noting that only about one-third of adults in the region meet basic financial literacy standards. Furthermore, the International Cooperative Research Group (2022) points to misinformation and a lack of awareness as primary barriers to cooperative growth. While leaders must naturally assess growth and operational health (Jamaluddin et al., 2023), the downstream influence of the financially literate member on these metrics is often overlooked. Financially savvy members are better equipped to avoid debt traps and contribute to the cooperative's capital (Ermawati et al., 2019), yet the direct link between this knowledge base and institutional performance remains under-researched. For example, Mwatondo and Wekesa (2020) found that literacy levels alone failed to fully explain growth dynamics in certain financial sectors, and Pratiwi et al. (2020) called for a deeper investigation into how member-level knowledge translates into institutional performance. Empirical research has yet to fully map how individual member competencies translate into institutional success within the local context of Bohol.

Methodology

Research Design

This study employed a quantitative research design to examine the relationship between cooperative members' financial literacy and the financial performance of large cooperatives. The researchers applied specific measures to assess both the members' financial literacy levels and the cooperatives' financial performance. Data were obtained through the distribution of structured questionnaires and the collection of financial statements. The gathered data were then systematically described and interpreted to derive substantive conclusions.

Research Environment

The research was conducted in Tagbilaran City, Bohol, Philippines, focusing on selected multi-purpose cooperatives. These cooperatives play an essential role in supporting local livelihoods and contributing to the province's economic development. Tagbilaran City is home to a well-established network of cooperatives across various sectors, making it an ideal setting for this study. According to the CDA, Tagbilaran City has 71 registered cooperatives, 43 of which are multi-purpose cooperatives located in the City of Tagbilaran, Bohol. As the capital of the province, Tagbilaran serves as the economic hub of Bohol, playing a significant role in local trade, business, and community development.

Research Instrument

The study utilized a structured questionnaire and financial ratio analysis as the primary research instruments to assess the financial literacy of cooperative members. The questionnaire consisted of 30 items equally distributed among three foundational constructs: financial knowledge, financial attitude, and financial behavior (10 items per dimension). The instrument was adapted from validated global instruments and structured in line with the OECD/INFE (2023) guidelines to ensure cross-contextual relevance and standardized measurement. Financial knowledge included both objective components (measured via multiple-choice and mathematical problem-solving items scored as correct/incorrect) and subjective components. Subjective knowledge, financial attitude, and financial behavior were assessed using a uniform four-point Likert scale ranging from strongly disagree to strongly agree. To establish the instrument's validity and reliability, a pilot test was conducted prior to data collection. The data yielded acceptable Cronbach's Alpha reliability coefficients above the standard 0.70 threshold for all subscales, confirming strong internal consistency.

The study also involved document analysis, specifically focusing on the cooperative's financial statements for the year 2024. These documents, along with related records and reports, were reviewed to extract relevant data. To evaluate the extent of financial performance, the researchers utilized financial ratios based on the STEPS (Stability, Turnover, Efficiency, Profitability, and Structure of Assets) method established by the Cooperative Development Authority (2021). These ratios provided the necessary financial data and allowed for the calculation of the selected indicators. The results were then used to determine the adjectival rating of the cooperative's financial performance.

Research Participants

The study involved members of multi-purpose cooperatives in Tagbilaran City, selected through purposive sampling. This non-probability method was used to target cooperatives that met specific study requirements, particularly those with established systems, larger memberships, and complex financial operations. Cooperatives were chosen based on three criteria: (1) being classified as large, (2) having a headquarters or branch in Tagbilaran City, and (3) operating as a multi-purpose cooperative. From a PCDO-provided list of qualified cooperatives, five were identified, and four agreed to participate.

After selecting the cooperatives, individual respondents were chosen using eligibility criteria to ensure relevance and accessibility. Respondents had to be active, verified members of a registered multi-purpose cooperative in Tagbilaran City. They also needed to be of legal age and have sufficient membership duration to be familiar with the cooperative's financial programs and operations. Purposive selection ensured that participants could provide informed, reliable data aligned with the study's objectives.

The sample size was justified based on standard rules for multiple linear regression and Ordinary Least Squares (OLS) path modeling. This framework dictates that for a model evaluating up to three concurrent predictor pathways, a minimum sample size of 74 is required to achieve adequate statistical power. The selection of 100 respondents exceeds this baseline, ensuring sufficient power to detect medium effect sizes. The sample was evenly divided among the four cooperatives, with 25 respondents each.

Data Analysis

Data were processed and analyzed using IBM SPSS Statistics and PROCESS Model. Cooperative financial performance was evaluated through document analysis of the 2024 audited financial statements to calculate ratios based on the STEPS method. To reconcile data across two distinct units of analysis, the STEPS performance metrics of the four cooperatives were disaggregated and assigned directly to the profile rows of their respective twenty-five individual member respondents. To evaluate the underlying mechanisms of member competencies, an Ordinary Least Squares (OLS) regression path analysis was performed to test the mediation model, specifically examining whether financial attitude mediates the relationship between financial knowledge and financial behavior. This mediation analysis utilized non-parametric bootstrapping with 5,000 resamples. Finally, simple linear regression was applied to the disaggregated data matrix to determine whether members' individual financial literacy indices significantly predict the overall organizational financial performance of their respective cooperatives.

Ethical Consideration

The study was conducted in strict accordance with ethical standards for research involving human participants. The Holy Name University Ethical Review Board granted formal ethical approval. The informed consent process was initiated prior to any data collection. During the recruitment phase, each potential participant was provided with a written consent form that detailed the study's objectives, the voluntary nature of their participation, and their right to withdraw at any time without penalty. Consent was obtained in a one-on-one setting to ensure participants could ask questions and fully understand the scope of the research. Regarding data obtained from cooperative records, the researchers secured formal authorization from the board of directors of each participating cooperative. This included specific permission to access and utilize institutional financial performance data for research purposes only. To protect confidentiality, all participants were anonymized using alphanumeric coding. All digital data, including survey responses and financial records, was stored in encrypted, password-protected files accessible only to the research team.

Results and Discussion

Profile of Respondents

Table 1 presents the demographic profile of the cooperative members. The findings reveal that the majority of the respondents are female, belonging mainly to the young and middle-aged adult groups, and many are college graduates. Furthermore, the members are generally classified under the lower-income (but not poor) group, according to the latest income classification by the Philippine Institute for Development Studies (PIDS), with only a few falling under the higher-income category.

Table 1. *Distribution of Respondents*

Profile	Options	Frequency
Age	18-24	9
	25-34	29
	35-44	29
	45-54	19
	55-64 above	14
	Total	100
Sex	Male	33
	Female	67
	Total	100
Education	High School	1
	High School Graduate	1
	College Level	11
	College Graduate	73
	Master's Degree	10
	Doctoral/PhD	1
	Vocational/Technical	3
	Total	100
Income	Less than ₱12,030	5
	₱12,030 - ₱24,059	53
	₱24,060 - ₱48,119	32
	₱48,120 - ₱84,209	7
	₱84,210 - ₱144,359	2
	₱144,360 - ₱240,599	0
	At least ₱240,600	1
	Total	100

Financial Literacy of Cooperative Members

The results show that cooperative members in Tagbilaran City have a generally high level of financial literacy, with a combined Financial Literacy Index (FLI) mean score of 3.50, interpreted as High. These findings are consistent with Perez & Lopez (2020), who found that school-based cooperative members demonstrated good financial understanding yet still engaged in problematic borrowing and lacked strong financial discipline. Quibra (2024) also observed that members of Sustainable Livelihood Program Associations had moderate financial knowledge and attitude but continued to show inconsistent financial behaviors, which limited their overall financial well-being. Together, these studies highlight the same pattern seen in our respondents: individuals may understand sound financial concepts, but consistently translating that knowledge into responsible financial behavior remains a challenge.

Table 2. *Level of Financial Literacy*

Component	Mean	Qualitative Interpretation
Financial Knowledge	3.57	High
Financial Attitude	3.60	High
Financial Behavior	3.33	High
Overall	3.50	High

Note. 3.26 - 4.00 High; 2.51 - 3.25 Moderate; 1.76 - 2.50 Low; 1.00 - 1.75 Very Low

Relationship Between Financial Literacy Constructs

The results reveal a strong and statistically significant relationship between financial knowledge and financial attitude ($\beta = 0.525$, $p < .001$), and between financial attitude and financial behavior ($\beta = 0.592$, $p < .001$). These beta values indicate strong positive effects, meaning increases in knowledge substantially improve attitude, and increases in attitude significantly enhance behavior. In contrast, the direct effect of financial knowledge on financial behavior was negligible and not statistically significant ($\beta = 0.024$, $p = .797$), showing that knowledge alone does not directly predict members' financial actions. The model further shows that financial knowledge accounts for 27.6% of the variance in financial attitude, while the combined effects of knowledge and attitude explain 36.7% of the variance in financial behavior, which are considerable proportions in behavioral research. These findings highlight that attitude acts as the key mechanism through which knowledge influences behavior. This shows a direct significant relationship between financial knowledge and financial attitude, which supports the findings of Shih et al. (2022). Their study explains that individuals with higher financial knowledge are more likely to develop positive financial attitudes. The study also found a direct significant relationship between

financial behavior and financial attitude. This is similar to the findings of Qisthi and Damayanti (2022), who confirmed a strong connection between these variables.

Table 3. Direct Effects

Dependent Variable	Predictor Variable	Std. Estimate	Std. Error	t-statistic	p-value	Interpretation
FA	FK	0.525	0.067	7.84	<.001	FK significantly increase FA
FB	FA	0.592	0.080	7.40	<.001	FA significantly increase FB
FB	FK	0.024	0.093	0.26	0.797	FK has no significant direct effect on FB

Note. FK – Financial Knowledge; FA – Financial Attitude; FB – Financial Behavior

Table 4 reveals that financial attitude fully mediates the effect of financial knowledge on financial behavior. This means that the influence of financial knowledge on behavior occurs entirely through its effect on attitude. The size of the indirect effect (0.311) indicates a moderately strong mediation, reinforcing that members who better understand financial concepts exhibit improved financial behavior only when this knowledge enhances their attitudes. This is in consonance with the results of Shrestha et al. (2023), which support the result that there is no significant direct relationship between financial knowledge and financial behavior. This suggests that financial knowledge alone is not enough to influence behavior, since its effect is mostly channeled through financial attitude. Overall, these findings indicate that financial knowledge, attitude, and behavior are closely interconnected. As highlighted by Yeolencia and Lestari (2024), financial knowledge positively and significantly influences financial attitudes and indirectly affects financial behavior through financial attitude, underscoring the mediating role of attitude in the relationship between financial knowledge and financial management behavior.

This implies that strengthening members' financial attitude, such as their motivation, confidence, and willingness to manage money responsibly, is crucial for translating knowledge into actual practices such as timely loan repayment, budgeting, and saving.

Table 4. Indirect Effects

Path	Point Estimate	Boot SE	Lower 95% CI	Upper 95% CI	Interpretation
FK → FA → FB	0.311	0.065	0.194	0.452	Full Mediation

Note. FK – Financial Knowledge; FA – Financial Attitude; FB – Financial Behavior; CI – Confidence Interval. Values are based on 5,000 bootstrap samples.

Financial Performance of Cooperatives

Table 5 presents the financial performance of the cooperative based on the CDA standards. The results show that, although the scores differ across components, the overall total performance of most cooperatives falls within the CDA's range for fair performance. This suggests that while the cooperatives have strengths in certain areas, their combined financial outcomes remain moderate. Comparable conclusions were highlighted in the study of Muryani et al. (2022), which stated that cooperatives with moderate member participation often achieve intermediate performance due to only partial optimization of member-driven activities.

Table 5. Financial Performance of Cooperatives

Cooperative	Stability	Turnover Ratio	Efficiency	Profitability	Structure of Assets	Total Score	Interpretation
A	14	2	30	7	17	70	Fair
B	15	1	23	12	13	64	Fair
C	14	3	27	7	9	60	Fair
D	14.5	2	21.5	8	10	56	Needs Improvement

Note. 91 – 100 Excellent; 81 – 90 Very Satisfactory; 71 – 80 Satisfactory; 61 – 70 Fair; <60 Needs Improvement

Relationship Between Financial Literacy and Financial Performance

The regression analysis reveals that financial literacy does not significantly predict cooperative financial performance, as shown by a non-significant beta coefficient ($\beta = 0.048$, $p = .632$). The extremely low coefficient indicates that a one-unit increase in financial literacy is associated with only a 0.048-unit increase in financial performance, which is statistically trivial. The model's $R^2 = 0.002$ means that financial literacy explains only 0.2% of the variance in financial performance, far below the threshold of practical significance. This implies that in these cooperatives, financial performance is determined by other organizational factors such as governance quality, managerial decisions, internal controls, and capital management, rather than the financial literacy levels of individual members. Thus, improving members' financial literacy alone is not enough to enhance the cooperatives' financial outcomes.

Yeolencia and Lestari (2024) found that financial literacy has no significant direct effect on organizational performance, similar to the non-significant result in this research. Ibitomi et al. (2024) also reported only a very small influence of financial literacy on SME performance, which reflects the extremely low beta coefficient observed in this study. Alkavi et al. (2025) further emphasized that financial literacy affects performance only when combined with mediating factors, such as financial behavior, financial inclusion, or technology use. Lusardi (2019) also noted that financial knowledge alone rarely improves financial outcomes without supportive organizational structures.

Table 6. *Regression Analysis*

Predictor	R ²	Std Estimate	Std Error	p-value	Interpretation
Financial Literacy	0.002	0.048	0.163	0.632	Not significant

Conclusion

This finding reveals that financial attitude serves as a full mediator between financial knowledge and financial behavior, addressing the crucial underlying mechanism that translates what an individual knows into how they actually act. While it is commonly assumed that simply teaching people financial facts will automatically improve their habits, our analysis demonstrates that financial knowledge lacks a direct pipeline to behavioral outcomes; instead, raw information remains functionally dormant unless it successfully alters an individual's internal mindset and psychological disposition toward managing money. Consequently, we can conclude that knowledge alone is a necessary but insufficient condition for behavioral change, as a person may fully comprehend the basics of finance but it will fail to yield better financial choices if their attitude remains unchanged.

Moreover, members' financial literacy does not influence the financial performance of the cooperatives. While members remain essential to cooperative operations, since they serve as the primary source of capital through share contributions and patronage, generate revenue through their patronage, and sustain activities through continuous transactions, their level of financial literacy alone is not enough to have an effect on the cooperative's performance. Instead, the cooperative's performance may be shaped more strongly by organizational factors such as governance, operations, and management. As a result, improving the financial literacy of members on its own does not lead to measurable enhancements of performance instead, cooperative performance depends more on how effectively the organization is managed and how actively members participate in its systems and operations.

As a recommendation, cooperative leaders and educators should move away from relying on information-heavy lectures about financial management and instead design training programs that connect with people on a personal level to shape their mindsets, confidence, and long-term financial attitudes. At the same time, because members' financial literacy alone does not automatically translate into financial performance, leaders must also invest in other aspects like improving governance and management, strengthening operations, and building user-friendly systems that make it easy and rewarding for members to participate regularly. Future researchers may conduct experiments to test whether attitude-focused training actually changes behavior better than traditional methods. Additionally, they may also look into specific cooperative aspects such as governance, operational workflows, and management systems as predictors of financial performance.

Contributions of Authors

Authors 1, 2, 3, and 4: conceptualization, data gathering, data analysis

Author 5: academic mentoring, research supervision, methodological guidance, manuscript review

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Conflict of Interests

The authors declare no conflict of interest.

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